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Reflecting on the Impact of General Education Electives in Engineering

In engineering, it is paramount to consider all factors involved in a problem before designing a solution. When considering these factors, background information, empathy, and collaboration with community members are all helpful tools to find a solution that positively impacts those affected. In my college education with general education electives, Iowa State has allowed me to deepen my understanding of the world around me to strengthen these tools and make my future solutions in engineering economically, globally, and socially viable. An engineer who merely considers the technical aspects of a problem may create a solution that works mathematically and scientifically but may be expensive to implement, create environmental sustainability concerns, or harm a particular population where the solution is implemented. The technical core classes of my degree don't necessarily teach me all of the impacts my engineering solutions will have or how to look ahead at all of the variables involved, so general education electives help bridge this gap and create learning opportunities to understand how our world works and what other perspectives people have that I may not.

In ECON 102 Macroeconomics, I learned the economic principles that govern our society. From the idea that everything is a trade-off and scarcity creates supply and demand, I quickly learned to apply these concepts to my design choices. Choosing whether to make a design energy efficient or highly performant creates a tradeoff between cost and productivity, and the price of a design may affect its supply of producers manufacturing the design and demand from consumers looking to buy the product. This context of how consumers and producers approach buying, selling, and trading sheds light on how businesses make decisions that affect the engineering of products and what consumers look for when deciding what engineering to invest in. This class has taught me to consider monetary and consumer perspectives when engineering.

Next, in ARCH 321, History of the American City, I learned how American engineers constructed cities across our country starting from when the U.S. was first founded. While I'm not a construction or civil engineer, I still learned a plethora of lessons from the successes, mistakes, and long-term effects of decisions made by policymakers and engineers early on (red-lining, government-subsidized housing, walking cities vs suburbs, cars and interstates, malls, etc.). Humanity can learn a lot and prevent future mistakes by observing the lessons we have already endured in history. In addition to the historical content, this class gave me a specific lens into the U.S.'s previous engineering mindsets and how they positively and negatively affected people regarding civil rights, quality of life, and cost of living. While the scale and technicality of my solutions may differ from constructing large cities, the considerations of groups of people, the economic facts such as demand and cost of goods sold, and the sustainability impacts are still very relevant to the engineering I will partake in my career.

Finally, in C E 388 Sustainable Engineering and International Development, this class combined economics and historical lessons, a technical understanding of energy consumption and sustainability, and how to develop solutions in regions worldwide. I learned from actual case studies of solutions for global challenges and how they impact parts of the world differently. One solution for a problem in the U.S. may benefit differently from another region due to limited materials and resources, operation costs, and cultural differences. Additionally, with standards like LEED and concepts in renewable agriculture, energy, and construction, I got a glimpse of all the materials and processes that make engineering solutions work and how changing energy sources can positively and negatively impact the environment. For my degree specifically, it's essential to understand where our energy comes from (for electronics and software especially) and how conserving it can make for lesser environmental impact. Overall, this class helped solidify international and ecological perspectives when building solutions to problems.

Thanks to a diverse group of classes, Iowa State has given me a broad perspective and breadth of background on humanity, both in the U.S. and internationally, which allows me to make more informed decisions, collaborate with diverse teams of people effectively, and consider all impacts of a solution to a

problem. With these tools and my technical knowledge, I am confident the solutions I implement are indeed solutions that positively impact the people they serve.